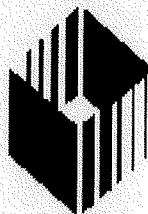
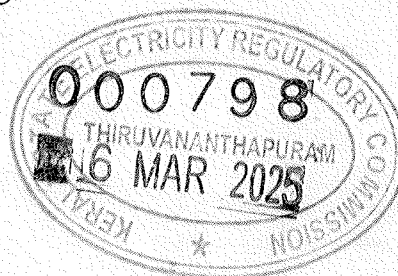


**PETITION FOR THE CAPITAL INVESTMENT
APPROVAL FOR THE IMPLEMENTATION OF
500 kW/ 1 MWh BATTERY ENERGY STORAGE
SYSTEM (BESS) ALONG WITH 400 kWp SOLAR
PLANT**

Submitted to
Hon. Kerala State Electricity Regulatory Commission

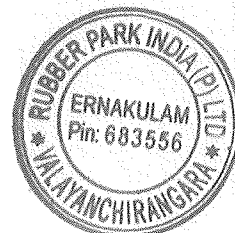
04th March 2025



RUBBER PARK INDIA PRIVATE LIMITED
(A JV of KINFRA & Rubber Board)

Regd. Off- 2 A, "Kautileeyam", Rubber Park, Valayanchirangara, P.O.
Ernakulam, Kerala, PIN-683 556.

Tele Phone : (0484) 2657218/ 2655548, FAX : (0484) 2655548,
E mail : rubberpark@vsnl.com, Web site: www.rubberparkindia.org.



BEFORE THE KERALA STATE ELECTRICITY REGULATORY COMMISSION

Filing No. /2025

Case No.

IN THE MATTER OF: Petition for the Capital Investment approval for the Implementation of 500 kW/ 1 MWh Battery Energy Storage System (BESS) along with 400 kWp Solar Plant.

And

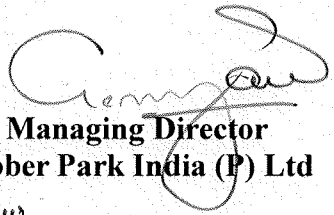
IN THE MATTER OF: Rubber Park India Private Ltd,
2 A Kauteeliyam,
Valayanchirangara,
Ernakulam -683 556

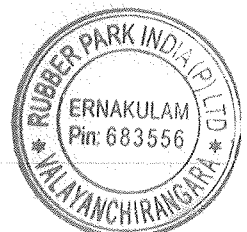
The petitioner named above respectfully submits as under:

1. The petitioner is distribution licensee in the areas as notified by the Government of Kerala vide G.O. (P) No.20/2003/PD dated 17-06-2003 (Gazette Extra Ordinary No.1199 dated 05-07-2003).
2. The Petitioner submits that the Hon'ble Commission has issued the following regulations namely, KSERC (Terms and Conditions for Determination of Tariff) Regulations, 2021 (hereinafter referred as "**KSERC MYT Regulations, 2021**").
3. As per the KSERC MYT Regulations 2021, petition shall be submitted for capital investment schemes exceeding Rs. 15 lakhs, along with comprehensive Detailed Project Reports, for the review and prior approval of Hon. Commission.
4. Accordingly, Rubber Park is filing this petition for the Capital Investment approval for the Implementation of 500 kW/ 1 MWh Battery Energy Storage System (BESS) along with 400 kWp Solar Plant.

Date: 04th March, 2025

Petitioner


Managing Director
Rubber Park India (P) Ltd



BEFORE THE KERALA STATE ELECTRICITY REGULATORY COMMISSION
C.V.Raman Pillai Road,Vellayambalam,Thiruvananthapuram-695010

Filing No.

Case No.

IN THE MATTER OF: Petition for the Capital Investment approval for the Implementation of 500 kW/ 1 MWh Battery Energy Storage System (BESS) along with 400 kWp Solar Plant.

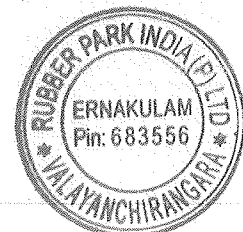
And

IN THE MATTER OF: Rubber Park India Private Ltd, Ernakulam.

Applicant


Managing Director
Rubber Park India Private Ltd,
Ernakulam-683556



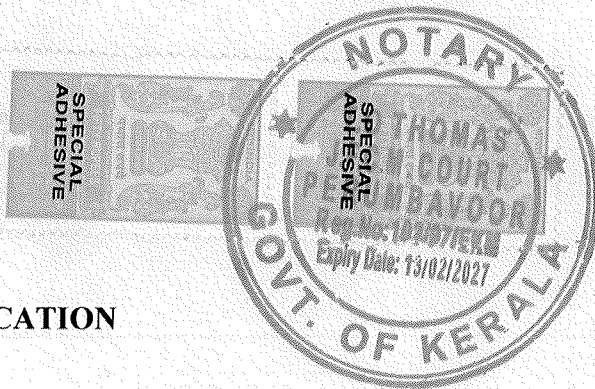


AFFIDAVIT VERIFYING THE APPLICATION ACCOMPANYING FILING OF PETITION FOR THE CAPITAL INVESTMENT APPROVAL FOR THE IMPLEMENTATION OF 500 KW/ 1 MWH BATTERY ENERGY STORAGE SYSTEM (BESS) ALONG WITH 400 KWP SOLAR PLANT.

I, **George V James**, S/o **V M James** (late) aged 61 years residing at **Vadakkumparambil House, Nandanath Kochacko Road, Thammanam, Ernakulam-682032** do solemnly affirm and state as follows:

1. I am the Managing Director of Rubber Park India (P) Ltd, 2 A, Kautileeyam, Valayanchirangara-683556, Ernakulam District, Kerala State and the petitioner in the above matter and I am duly authorized by the company to make this affidavit on its behalf. I solemnly affirm at Perumbavoor on this 04th day of March 2025 that

- (i). The contents of the above petition are true to my information, knowledge and belief. I believe that no part of it is false and no material has been concealed there from
- (ii). The statements made in paragraphs of the accompanying application herein now shown to me are true to my knowledge and are derived from the official records made available to me and are based on information and advice received which I believe to be true and correct.



Deponent

**Managing Director
Rubber Park India Private Ltd,
Ernakulam-683556**

VERIFICATION

I, the above-named deponent solemnly affirm at Perumbavoor on this, the 04th day of March 2025 that the contents of the affidavit are true to my information, knowledge and belief, that no part of it is false and that no material has been concealed there from.

Deponent

**Managing Director
Rubber Park India Private Ltd,
Ernakulam-683556**

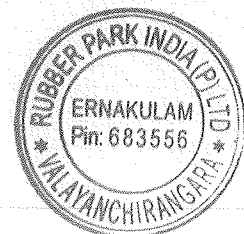
Solemnly affirmed and signed before me
Advocate and Notary

T.O. THOMAS, B. A. LL.B.
THACHIL
ADVOCATE & NOTARY
(GOVT. OF KERALA)
PERUMBAVOOR
INDIA



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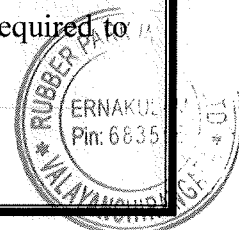
1. BACKGROUND

The Rubber Park India (P) Ltd is a joint venture of Kerala Industrial Infrastructure Development Corporation (Kinfra) and Rubber Board. The Company was incorporated in 1997 under The Companies Act 1956 for establishing industrial parks, exclusively for rubber and rubber wood-based industries with world class infrastructure. The first park was set up in 110 Acres at Irapuram in Kunnathunadu Taluk, near Perumbavoor, in Ernakulam Dist. The first park was established on a sprawling 110-acre of land in Irapuram village, situated within the Kunnathunadu Taluk, close to Perumbavoor, in the Ernakulam District. The land within the Irapuram park has been almost entirely allocated to a diverse array of entrepreneurs, with 51 industrial units already in full operational swing on the allotted land. The second Rubber Park has been established on a 20-acre plot of land in Piravanthoor, within the Pathanapuram Taluk, in close proximity to Punalur within the Kollam District and has commenced the allocation of land to aspiring entrepreneurs for the purpose of establishing various industrial ventures.

Rubber Park India Pvt. Ltd. (RPIPL) is a small DISCOM in Kerala. The Government of Kerala granted it a license for power distribution at Rubber Park, Irapuram, vide G.O. (P) No. 20/2003/PD dated 17-06-2003, as notified in Gazette Extra Ordinary No. 1199 dated 05-07-2003. Rubber Park owns a 110/11 kV, 2X12.5 MVA capacity EHT Substation and well-established power distribution network for supply of power to various consumers in the park. The infrastructure includes two 11 kV ring mains and two dedicated feeders from the substation, ensuring a reliable and high-quality power supply to the consumers. Rubber Park has implemented a prepaid billing system for its consumers, ensuring transparent and efficient billing processes. In addition to serving the consumers inside the park, provided open access facility to one of the consumers of KSEB Ltd at 11 kV voltage level.

2. Objective of the Capital Investment

The Hon'ble Kerala State Electricity Regulatory Commission (KSERC) notified the KSERC (Renewable Energy and Net Metering) (Second Amendment) Regulations, 2024, on 07.08.2024. Under this amendment, KSERC has mandated the Energy Storage Obligation (ESO) starting from FY 2025-26, in addition to the existing Renewable Power Obligation (RPO). In the ARR&ERC order for FY 2022-27, KSERC directed KSEB Ltd. to meet the RPO of RPIPL, with RPIPL being required to



reimburse the additional cost to KSEB Ltd. at rates determined by the Commission from time to time. However, with the recent regulatory amendment, RPIPL must comply with the ESO from FY 2025-26.

The Government of India has set ambitious targets of achieving 500 GW of renewable energy capacity by 2030 and net-zero emissions by 2070. Likewise, the Government of Kerala has outlined its roadmap for achieving 100% renewable energy by 2040 and net-zero emissions by 2050. In alignment with these national and state-level commitments, RPIPL is taking proactive steps to integrate renewable energy sources and advanced energy storage solutions within its infrastructure.

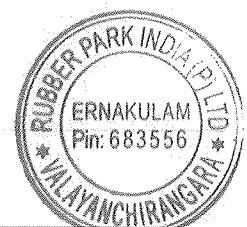
In order to fulfil this requirement, RPIPL is proposing to implement a 500 kW/1 MWh Battery Energy Storage System (BESS) along with a 400 kWp solar plant at Rubber Park, Irapuram. The primary objective of this project is to harness solar energy during the daytime and store it efficiently for use during peak demand hours.

The integration of BESS with the solar plant is expected to improve the reliability of power supply, stabilize fluctuations in renewable energy generation, and provide a sustainable alternative to conventional electricity consumption. Furthermore, this initiative will contribute to reducing the park's overall carbon footprint, in line with Kerala's broader sustainability goals.

3. Project Cost Estimate

RPIPL invited a tender in September 2024 for the design, supply, installation, testing, and commissioning of a 500 kW/1 MWh BESS at Rubber Park, Irapuram. The tender for the 400 kWp solar plant is under preparation. The scope of work includes:

- Deployment of 1x500kW / 1000kWh BESS at Rubber Park-Irapuram (Ernakulam Dt.)
- Site survey, design, engineering, planning, manufacturing, supply, transportation, and installation
- Erection, commissioning, and testing, including labor and supervision
- Establishment of grid connectivity at the 11 kV Grid Point
- Associated civil and fabrication works



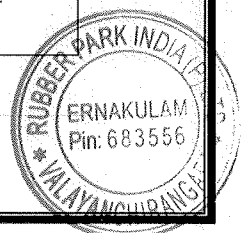
- EMS system and its integration with KSEBL SCADA System
- Supply of mandatory spares
- Technical support for the contract period.

Eight bidders participated in the first tender; however, due to the insufficient number of technically qualified bidders, the tender was cancelled. A second tender was invited in December 2024 with revised conditions based on some of the clarifications from the evaluation of the first tender, with the following major conditions:

- The contract period for the BESS was revised from **10 years to 5 years**, as some bidders quoted battery replacement in 5 years and there was no clarity on the battery life beyond this period. Additionally, battery storage technology is rapidly evolving, with ongoing research and development in the field. This presents a high likelihood of more cost-effective and advanced solutions emerging within the next 5 years. Consequently, required capacity of the BESS was also revised to 500 kW / 1000 kWh at the end of the 5th year.
- The procedure for consideration of the **performance certificate for works executed abroad** was detailed to provide more clarity.

In the second tender 10 bidders were participated. The price bids of the four technically qualified bidders were opened on 13.02.2025. The price bid opening summary is as follows:

Sl. No.	Name of Bidder	Amount Quoted (INR)	Bid Rank
1	Hykon India Ltd	2,49,89,800.00/-	L-1
2	M/s. Pacemaker Solutions	2,65,00,000.00/-	L-2
3	EPSL Trigenation (P) Ltd	3,70,00,000.00/-	L-3
4	MIDCOSE	3,98,00,000.00/-	L-4



The total cost for the implementation of BESS along with solar plant is Rs. 4.35 Crores (inclusive of GST), comprising Rs. 2.95 Crores for the BESS and Rs. 1.4 Crores (estimated) for the solar power plant. The tender for the 400 kWp grid connected solar plant is under preparation.

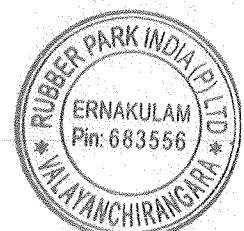
Meanwhile, RPIPL has requested financial assistance in the form of a grant from the Ministry of Power, Government of India, and Energy Management Centre of Kerala.

4. Technical Justification

The lowest bidder has proposed a designed gross battery capacity of 1671.99 kWh at the first year and 1510.95 kWh at the end of fifth year. The year wise battery capacity and total AC usable capacity as per the technical details submitted by the lowest bidder is furnished as below.

Year	Gross battery capacity (kWh)	AC usable capacity (kWh)
First	1671.99	1408
Second	1630.19	1365
Third	1589.43	1330
Fourth	1549.70	1296
Fifth	1510.95	1263

As per the technical specifications, the required BESS capacity was 500 kW / 1000 kWh at the end of the 5th year. The lowest bidder proposed a gross battery capacity of 1671.99 kWh, factoring in degradation and losses to ensure an AC usable capacity of 1000 kWh at the end of the 5th year. The BESS is sufficient to meet the ESO up to the year 2028-29. The year wise energy storage obligation as per the regulation of Hon. KSERC and the corresponding year wise output from the BESS is detailed below.

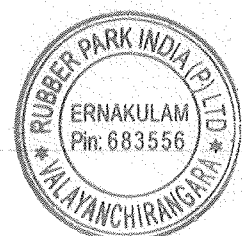


Year	Projected Annual Energy Sales (kWh)	ESO (%)	ESO (kWh)	BESS Output (kWh)
2025-26	3,58,00,000	0.15	53,700	5,13,920
2026-27	3,68,00,000	0.5	1,84,000	4,98,225
2027-28	3,77,00,000	0.75	2,82,750	4,85,450
2028-29	3,86,00,000	1	3,86,000	4,73,040
2029-30	3,95,00,000	2	7,90,000	4,60,995

The usable capacity quoted by the lowest bidder at the end of the 5th year is 1263 kWh. This ensures that RPIPL will continue to receive the required output of 1000 kWh from the BESS during the 6th and 7th years as well, despite the defect liability period with the contractor being limited to 5 years as per the tender.

Furthermore, although the present capacity may not be sufficient to fully meet the Energy Storage Obligation (ESO) requirements for 2029-30, the BESS is expected to generate significantly higher output than the required ESO during the first four years of operation. This surplus energy utilization in the initial years will help optimize overall system performance.

Additionally, BESS technology is continuously evolving, with ongoing research and development in battery technology aimed at improving efficiency, durability, and cost-effectiveness. By the 5th year, advancements in battery variants and emerging cost-effective solutions are anticipated, potentially offering more viable options for future capacity augmentation.



5. Project Time line

The project is scheduled to be completed by 7 months from the date of award. The tender was already finalised and the work can be awarded immediately after the necessary approvals. The time line of the project is detailed below.

Project Timeline: 500 kW/1 MWh BESS with 400 kWp Solar Plant

Activity Description	Duration	Month 1	Month 2	Month 3	Month 4	Month 5	Month 6
Land Clearing	2 weeks	■					
Design	2 weeks	■					
Supply	2 months		■	■			
Installation	2 months			■	■		
Testing & Commissioning	2 months					■	■

6. Timing of the Investment

The timing of the investment in the **500 kW / 1000 kWh Battery Energy Storage System (BESS)** at Rubber Park, Irapuram, is **well-aligned with current and future energy storage requirements**, considering the following factors:

- **Energy Storage Obligation (ESO) Compliance**

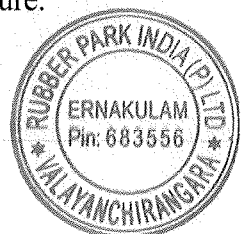
- The BESS is designed to meet ESO requirements as per Hon. KSERC regulations up to **2028-29**.
- During the initial years (**2025-26 to 2028-29**), the projected output from the BESS significantly exceeds the required ESO, ensuring compliance and surplus capacity for operational flexibility.
- Although the projected output in **2029-30 (4,60,995 kWh)** falls short of the required ESO (**7,90,000 kWh**), advancements in battery technology may provide cost-effective expansion options before then.

- **Technology Evolution and Cost Considerations**

- BESS technology is evolving rapidly, with ongoing R&D focused on improving efficiency, reducing degradation rates, and lowering costs.
- With significant advancements expected within the next 5 years, deferring investment beyond this period could provide access to more cost-effective and high-performance storage solutions.
- The **5-year contract period** ensures flexibility, allowing RPIPL to adapt to technological improvements and regulatory changes in the future.

- **Market Readiness and Bidder Participation**

- The **first tender** was cancelled due to insufficient qualified bidders.
- The **second tender (December 2024)** incorporated refined technical specifications and performance criteria, ensuring better clarity and competitiveness.
- Proceeding with the investment now ensures that RPIPL can secure a **commercially viable solution** while still leveraging advancements in energy storage in the near future.



- **Defect Liability and Performance Longevity**

- The lowest bidder has designed a **gross battery capacity of 1671.99 kWh**, with a usable capacity of **1263 kWh at the end of the 5th year**.
- RPIPL may have the opportunity to continue receiving the required **1000 kWh output during the 6th and 7th years** also, even beyond the **5-year defect liability period**.
- This buffer period provides additional flexibility for future expansion or replacement planning.

- **Projected Demand Growth vs. BESS Output**

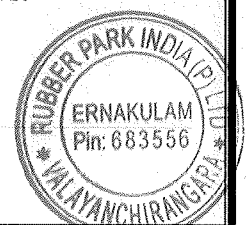
- The projected **annual energy sales** are expected to grow from **35.8 Million units in 2025-26 to 39.5 Million units in 2029-30**, an increase of **10.34% over 5 years**.
- The **ESO requirement** also grows proportionally, reaching **7,90,000 kWh in 2029-30**, which **exceeds the projected BESS output of 4,60,995 kWh** at that time.

- **Utilization of Surplus Capacity in Initial Years**

- In the first four years, the BESS output significantly exceeds the ESO requirements:
 - **2025-26:** 5,13,920 kWh (Surplus of 4,60,220 kWh)
 - **2026-27:** 4,98,225 kWh (Surplus of 3,14,225 kWh)
 - **2027-28:** 4,85,450 kWh (Surplus of 2,02,700 kWh)
 - **2028-29:** 4,73,040 kWh (Surplus of 87,040 kWh)
- This **initial surplus** can be leveraged for support other licensees on commercial terms, peak load management, or future regulatory adjustments as approved by the Hon. KSERC.

- **Flexibility for Future Expansion**

- Although the **BESS capacity may fall short of ESO obligations by 2029-30**, the evolving **battery market is expected to offer cost-effective solutions for capacity augmentation** by that time.
- The decision to implement a **5-year contract instead of 10 years** ensures that RPIPL is not locked into older technology, allowing room for future expansion as demand grows.



The investment timing is **optimal**, balancing regulatory compliance, market maturity, technological advancements, and cost efficiency. The 5-year contract structure provides flexibility while ensuring adequate performance and compliance in the medium term.

The investment is **aligned with short- to medium-term demand growth** and ESO requirements. While capacity falls short by 2029-30, **technological advancements and surplus output in earlier years provide strategic flexibility** for future scaling.

7. Mode of Funding for the Project

RPIPL has sought financial assistance from the **Energy Management Centre (EMC), Kerala**, to support the implementation of the **500 kW / 1000 kWh Grid-Connected Battery Energy Storage System (BESS)**. Given the capital-intensive nature of the project, RPIPL aims to secure funding at a **low-interest rate** to ensure financial viability and affordability.

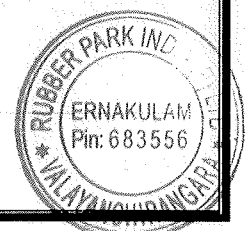
EMC has initially proposed the following loan terms for the project:

- **Fixed Yearly Interest Payment: 2%**
- **Administrative Charges: 5% (one-time).**

The **Effective Interest Rate of loan proposed by EMC is Approximately 5.65% per annum**, calculated on an outstanding principal balance basis. Accordingly, RPIPL has **requested a revision of the loan terms** to improve financial feasibility and reduce the total cost of borrowing. The revised proposal from RPIPL includes:

- **Interest Rate: 2% per annum** on the outstanding principal balance
- **Administrative Charges: 1% (inclusive of GST)**
- **Loan Tenure: 5 years**
- **Funding Mode: 100% loan financing from EMC**

RPIPL intends to **proceed with project implementation under a 100% loan financing model** from EMC, subject to finalization of the funding terms. The Full loan financing ensures that RPIPL does



not have to allocate significant internal funds, allowing capital to be preserved for other critical infrastructure projects.

- **Affordable Financing:** The requested interest rate of **2%** with **1% administrative charges** is competitive compared to market lending rates, making the investment financially sustainable.
- **Encouraging Renewable Energy Initiatives:** The concessional financing from EMC aligns with the government's objective of promoting **energy storage solutions** and integrating renewable energy into the grid.
- **Cost Optimization:** By securing a low-interest loan, RPIPL can ensure that the cost of the BESS project remains within a **reasonable financial framework** while ensuring compliance with regulatory mandates.

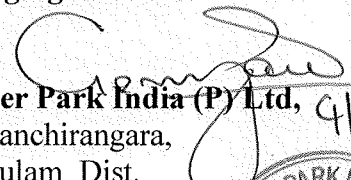
RPIPL is currently engaged in **negotiations with EMC** to finalize the loan terms. The objective is to secure the best possible financing arrangement, ensuring that the project remains financially viable while meeting **ESO obligations** and improving **power reliability at Rubber Park, Irapuram**.

8. PRAYER

Rubber Park humbly prays to the Hon'ble Commission that:

- (a) The accompanying petition for the Capital Investment approval for the implementation of a **500 kW/1 MWh Grid-Connected Battery Energy Storage System (BESS) along with a 400 kWp Solar Plant at Rubber Park, Irapuram**, with a total cost estimate of **Rs. 4,35,00,000**, may kindly be approved.
- (b) Approve the cost recovery of the proposed Capital Investment to ensure the financial sustainability of Power distribution business of RPIPL.

Managing Director


Rubber Park India (P) Ltd, 4/3/25
Valayanchirangara,
Ernakulam Dist.

